

AI Tutor Web App — Project Report

Project Overview

The **AI Tutor** is a smart education platform built for modern learners. Currently, it offers:

-  **AI-powered doubt solving**
-  **A curated Question Bank**
-  **Exam generation & evaluation system (under development)**
-  **Collaborative study space (under development)**
-  **Study planner with progress tracking (under development)**

All features are designed with student efficiency and personalization in mind.

Tech Stack Overview

Frontend (React + Vite)

- **React (v18.2)** – Building reusable and interactive UIs
- **Vite** – Lightning-fast development server and optimized production builds
- **Tailwind CSS** – Rapid styling and responsive design
- **React Router DOM** – SPA navigation
- **Framer Motion** – Smooth animations and transitions
- **React Toastify** – In-app notifications
- **React Markdown + KaTeX** – Markdown rendering with LaTeX math support

- **Socket.IO Client** – Real-time communication
 - **Firebase Auth** – Google login integration
 - **Lucide Icons & React Helmet** – Modern UI and SEO features
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Backend (Node.js + Express)

- **Express (v5.1)** – Core server framework
 - **MongoDB + Mongoose** – Database and ODM for user data, questions, planner, and chat
 - **JWT (jsonwebtoken)** – Secure user authentication
 - **Passport + Google OAuth** – Social login integration
 - **Socket.IO** – Real-time chat and live collaboration
 - **Cloudinary + Multer** – Image and file uploads
 - **@google/generative-ai** – Access to LLMs for chat and content generation
 - **Joi** – Input validation
 - **Helmet, CORS, XSS-Clean, Mongo-Sanitize, Rate Limiter** – Security middleware
 - **Dotenv** – Secure config management
 - **Morgan** – Request logger
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Dev Tools

- **Nodemon + Concurrently** – Hot reloading during development
- **ESLint + Prettier** – Code quality and consistency
- **Vite in backend** (for shared config with frontend)

Learning Outcomes

Technical Mastery

- Created a scalable **full-stack MERN application**
- Integrated **Socket.IO** for real-time features like live chat and collaboration
- Implemented **secure user authentication** with JWT and Google OAuth
- Built RESTful APIs with **Express + MongoDB**
- Handled file uploads securely with **Multer and Cloudinary**
- Used **OpenAI/Generative AI APIs** for intelligent content generation

Product Engineering

- Designed **modular architecture** for better maintainability
- Ensured **code security** using sanitizers and rate limiters

Project Skills

- Practiced **CI-friendly development** with nodemon, eslint, and prettier
- Gained experience in **real-world API design, frontend-backend integration**, and UX focus
- Built features with **future scalability and team collaboration in mind**

